



NONLINEAR MODEL

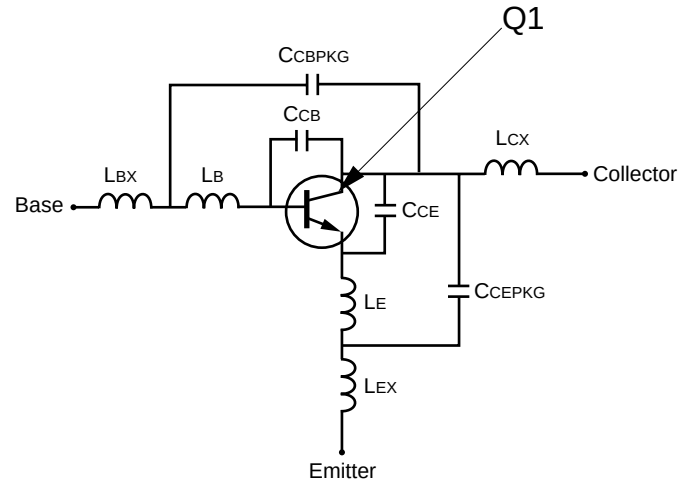
NE685M13

BJT NONLINEAR MODEL PARAMETERS (1)

Parameters	Q1	Parameters	Q1
IS	7e-16	MJC	0.34
BF	109	XCJC	0.7
NF	1	CJS	0
VAF	15	VJS	0.75
IKF	0.19	MJS	0
ISE	7.9e-13	FC	0.5
NE	2.19	TF	2.5e-12
BR	1	XTF	5.2
NR	1.08	VTF	4.58
VAR	12.4	ITF	0.011
IKR	0	PTF	0
ISC	0	TR	1e-9
NC	2	EG	1.11
RE	1.3	XTB	0
RB	10	XTI	3
RBM	8.34	KF	0
IRB	0.009	AF	1
RC	10		
CJE	0.4e-12		
VJE	0.812		
MJE	0.5		
CJC	0.18e-12		
VJC	0.75		

(1) Gummel-Poon Model

SCHEMATIC



ADDITIONAL PARAMETERS

Parameters	68533
CCB	0.1e-12
CCE	0.14e-12
LB	0.35e-9
LE	0.4e-9
CCBPKG	0.05e-12
CCEPKG	0.05e-12
LBX	0.05e-9
LCX	0.05e-9
LEX	0.05e-9

UNITS

Parameter	Units
time	seconds (S)
capacitance	farads (F)
inductance	henries (H)
resistance	ohms (Ω)
voltage	volts (V)
current	amps (A)

MODEL RANGE

Frequency: 0.1 to 4.0 GHz
Bias: $V_{CE} = 0.5\text{ V to }3\text{ V}$, $I_C = 0.5\text{ mA to }20\text{ mA}$
Date: 09/02

Life Support Applications

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